

On a Preliminary Model for Smartphone Battery Power Consumption

Summary

Smartphones are playing an increasingly important role in our daily lives. Although many past studies have established various models regarding individual components, people still lack a comprehensive understanding or theoretical model of their behaviors regarding power consumption. In this paper, we establish a comprehensive mathematical model for smartphone power consumption based on studies on individual components, and administered a simple prediction on the State-of-Charge (SOC) and remaining drainage time of a smartphone battery.

The outline of our research is to independently consider the behaviors of components, then combine them. We consider the following components: screen, processing units (CPU and GPU), network modules, speakers, peripherals (camera and microphone), and battery internal resistance loss. We consider solid-founded physical and chemical principles, such as the Arrhenius equation, Nernst equation, Fourier's law, and Newton's law of cooling, to establish our model.

Subsequently, we combine all those factors. In the past studies, various formulas were proposed, and we extract the relationships between variables, further examine the exact values of most undetermined constants, and form a complete model. Due to confidential reasons and technical limitations, we approximate some undetermined constants basing on typical values and real-world experience. In light of these, we conduct computer simulations to verify the accuracy of our model.

After a general simulation, we delete different factors from our program one by one to compare their contributions to the total power consumption. Next, we examine how the model changes under different changes in parameters. Finally, we provide practical recommendations for different entities to optimize battery performance and longevity.

Overall, through simplifying scenarios and establishing reasonable assumptions, we manage to build a model that can, to a certain extent, predict the SOC value and remaining drainage time of a smartphone battery, reflecting the current power consumption, and hence offer suggestions for improvement.

Keywords: Smartphone Battery, Power Consumption Model, State-of-Charge (SOC), Thermal Effects, Battery Degradation

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1 Introduction

1.1 Background

In our modern everyday life, lithium batteries have become an essential part of portable electronic devices, especially smartphones. As the performance of smartphones improves, so does their power consumption and our thirst for power. However, the battery capacity is always limited, thus the remaining battery energy have always been a concern for smartphone users.

However, this problem is not simple. Lithium-ion batteries, as the most commonly-used batteries in smartphones, exhibit high non-linearity and uncertainty in their discharging behaviors. Various factors, such as temperature, load conditions, and battery aging, can significantly affect the battery performance.

In light of this, establishing the mathematical model of smartphone battery consumption is of great significance. This model should incorporate various factors that influence the power consumption, such as screen usage, network activity, processor load, peripheral usage and so on. Moreover, the model should also take into account the battery degradation over time, which affects the overall battery life and performance. In pursuit of higher accuracy, this model, under ideal circumstances, should base entirely on physical and chemical principles. But in practice, due to the incomplete knowledge we are in control of, experimental data, statistical methods, and subsequently, empirical formulas are also necessary.

1.2 Restatement of the Problem

Our team is required to establish a mathematical model to calculate the State-of-Charge (SOC, in percentage) and the remaining time of a smartphone battery before it is completely drained. The whole project consists of five subtasks:

1. Develop a continuous-time model to predict the SOC value in percentage.
2. Use the mentioned model to predict the time-to-empty (the time before the battery is completely drained).
3. Use the mentioned model to compare the contribution of different factors.
4. Examine how the model changes under different assumptions, parameters, and usage patterns.
5. Use the mentioned model to provide practical recommendations for different entities to optimize battery performance and longevity.

Despite the restated problems being quite clear, we still follows another logic to organize the contents below.

1.3 Our Work

First of all, we establish a theoretical model to calculate the power consumption of different components in a smartphone, including the screen, processing units (CPU and GPU), network modules, speakers, and peripherals (camera and microphone). Then, we combine these parts to form the total

power consumption of the smartphone. Based on this, we can calculate the SOC value and the remaining drainage time.

Then, we used computer simulations to verify the accuracy of our theoretical model. After a general simulation, we deleted different factors from our program one by one to compare their contributions to the total power consumption. Next, we examined how the model changes under different assumptions, parameters, and usage patterns. Finally, we provided practical recommendations for different entities to optimize battery performance and longevity.

There might be inaccuracies, even fallacies in our model due to technical limitations, insufficient knowledge, and lack of experimental data. We have listed these limitations at the end of the paper. The strengths of our paper are also summarized there.

Here is a brief overview of our work, in the form of a flowchart.

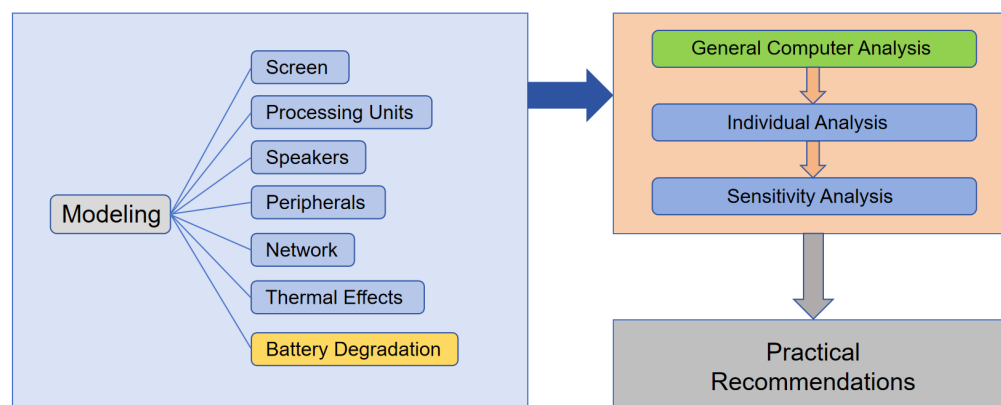


Figure 1: Flowchart of our Work

The assumptions and justifications we use are listed in the corresponding subsections respectively.

2 Notations

The notations for most variables we use in this paper are listed below. All variables are real numbers unless otherwise specified, here we do not include any indices or superscripts, which are further explained in the corresponding contexts. Also, these meanings may be restated in the contexts for the convenience of the reader. We use SI units unless specified.

Symbol	Meaning	Unit
(SOC)	State of charge	%
E	Energy / Battery capacity	J / Wh or Ah
P	Power	W
t	Time	s
T	Temperature	K
U	Voltage	V
$\mathcal{E}$	Electromotive force	V
I	Current	A
f	Frequency	Hz
K	Equilibrium constant	-
k	Rate constant / Thermal conductivity	- / W/(m·K)
Q	Reaction quotient / Heat	- / J
q	Heat transfer rate	W
A	Area	m ²
C	Capacitance	F
c	Concentration / Specific heat capacity	mol/m ³ / J/(kg·K)
m	Mass	kg
Z	Complex impedance	Ω

The scientific constants we use are listed below.

Symbol	Meaning
k_B	Boltzmann constant
R	Universal gas constant
F	Faraday constant

Unless marked by a pair of parentheses, the input value of all functions only include the first monomial (not separated by a dot product) or fraction after the function name (+/- sign included). For example, $\exp -a/b$ means $e^{-a/b}$, $\exp ab + c$ means $e^{ab} + c$ but not e^{ab+c} , and $\exp a/b \cdot c$ means $e^{\frac{a}{b}} \cdot c$ but not $e^{\frac{a}{b \cdot c}}$. Formats for fractions have also been adjusted for better readability; for example, $a/bc = \frac{a}{bc}$ is written instead as $a/(bc)$. Moreover, variable names with 2 or more letters are written within a pair of parentheses for better readability; for example, SOC is written as (SOC). The notation $\log$ denotes the natural logarithm.

3 Theoretical Model

3.1 Overview

Assumption 3.1-1 The remaining drainage time can be estimated using the average power consumption over the past period, from full charge to the current time ($t = 0$ to $t = t_0$).

Justification 3.1-1 By averaging the power consumption over a recent period, we can obtain a reliable estimate of the current usage pattern, which can be used to predict the remaining drainage time.

We generally consider the following factors: screen, processing units, network, speakers, peripherals (camera, microphone, etc.), and battery internal resistance loss. Thus, the total power consumption is given by

$$P_{\text{total}} = \frac{dE}{dt} \quad (1)$$

$$P_{\text{total}} = P_{\text{screen}} + P_{\text{processor}} + P_{\text{network}} + P_{\text{sound}} + P_{\text{peripherals}} + P_{\text{battery loss}} \quad (2)$$

The goal is to solve

$$(\text{SOC})_{t_0, N} = \left(\frac{(\text{Energy left in battery})}{E_N} \right) \times 100\% = \left(\frac{E_N - \int_0^{t_0} P_{\text{total}} dt}{E_N} \right) \times 100\% \quad (3)$$

and

$$t_{\text{remain}} = \frac{E_0 - \int_0^{t_0} P_{\text{total}} dt}{t_0^{-1} \int_0^{t_0} P_{\text{total}} dt}. \quad (4)$$

3.2 Screen

Assumption 3.2-1 We only consider the OLED screens.

Justification 3.2-1 OLED screens are widely used in modern smartphones due to their superior display quality compared to traditional LCD screens, which displays a trend of OLED becoming the mainstream.

Assumption 3.2-2 The thermally-stimulated migration rate is proportional to the carrier mobility.

Justification 3.2-2 The thermally-stimulated migration rate affects the movement of charge carriers within the OLED material, which in turn influences the carrier mobility. Under the condition of constant electric field and concentration of carriers, this proportionality can be assumed.

Assumption 3.2-3 The ambient lighting in one day follows a trigonometric function.

Justification 3.2-3 The ambient lighting varies periodically throughout the day, with peaks during daylight hours and troughs during nighttime. A trigonometric function can largely capture this periodic behavior.

Assumption 3.2-4 The brightness of the screen is proportional to the ambient lighting.

Justification 3.2-4 Users often adjust the screen brightness based on the surrounding ambient lighting to ensure optimal visibility and reduce eye strain. And for the proportionality, see work [1].

3.2.1 Basic Model

First of all, we need to consider the mechanism of OLED screens. OLED screens work by passing current through organic compounds that emit light when energized. The current consists of electrons and holes (we summarize them as "carriers") that recombine to produce photons. The efficiency of this process is influenced by temperature. Then we may construct our model as follows.

According to the definition of power, we have

$$P \propto \eta \cdot J \cdot U \quad (5)$$

where η is the efficiency (seen as a constant), J is the current density, and U is the voltage. Moreover, by Arrhenius equation, the carrier mobility, or the number of carriers whose energy exceeds the activation energy, is given by

$$\mu(T) = \mu_0 \exp -\frac{E_a}{k_B T} \quad (6)$$

where μ_0 is a constant, E_a is the activation energy, k_B is the Boltzmann constant, and T is the temperature in Kelvin. Thus, under stable illumination conditions when P , J are constants, we have

$$U \propto U_0 \exp \frac{E_a}{k_B T} \quad (7)$$

Now we may assume that the thermally-stimulated active migration rate k_{nr} is proportional to $\mu(T)$, i.e.

$$k_{nr} = k_0 \exp -\frac{E_a}{k_B T} \quad (8)$$

Lighting efficiency under high temperature

$$\eta = \frac{k_r}{k_r + k_{nr}} = \frac{1}{1 + A \exp -E_a/(k_B T)}, A = \frac{k_0}{k_r} \quad (9)$$

where k_r is the rate of electromagnetic radiation. Here we assume P , U are constants, so

$$J \propto \eta^{-1} = 1 + A \exp -E_a/(k_B T) \quad (10)$$

hence,

$$P_{\text{screen}} \propto \exp \frac{E_{a1}}{k_B T} \cdot \left(1 + A \exp -\frac{E_{a2}}{k_B T} \right) \quad (11)$$

3.2.2 Influence of Ambient Lighting

Citing the works from previous scholars^[2], we have

$$P_{\text{screen}} = a_0 + (Br)(a_R P_R + a_G P_G + a_B P_B) \quad (12)$$

where indices R, G, B denote red, green, and blue respectively; a_0 , a_R , a_G , a_B are constants related to the specific screen; P_R , P_G , P_B are the percentages of red, green, and blue respectively; and (Br) is the brightness of the screen.

According to Assumption 3.2-3 and 3.2-4, we have

$$(Br) = \varepsilon L_{\text{amb}} = \varepsilon L_0 \sin(\omega t + \delta) + L_{\text{base}} \quad (13)$$

where L_0 is the maximum ambient lighting in a day, δ is a phase shift depending on the geographical location, L_{base} is the base ambient lighting, and ε is the proportionality constant. When considering indoor usages, we may set ε to a smaller value, or even 0.

3.3 Network

Assumption 3.3-1 We only consider the data transmission power consumption.

Justification 3.3-1 Data transmission is the primary function of network modules in smartphones, and it significantly contributes to power consumption. Other functions, such as idle listening or background processes, generally consume less power and can be considered negligible for the purpose of this model.

Assumption 3.3-2 We see GPS usage as a type of network activity.

Justification 3.3-2 GPS modules in smartphones rely on network connectivity to receive satellite signals and provide location services. Therefore, the power consumption associated with GPS usage can be reasonably modeled as part of the overall network activity.

Citing the works from previous scholars^[3], we have the following equation

$$P_{\text{network}} = \sum_i \alpha_{ui} \theta_{ui} + \alpha_{di} \theta_{di} + \beta \quad (14)$$

where α_{ui} , α_{di} , β are constants related to the specific network type, and θ_{ui} , θ_{di} are the upload and download throughputs (in Mbps) respectively. The index i denote different processes.

3.4 Processing Units

Assumption 3.4-1 We consider only static and dynamic power consumption of the CPU.

Justification 3.4-1 Static and dynamic power consumption are the primary contributors to the overall power consumption of a CPU. Other factors, such as short-circuit power consumption and leakage power consumption, are generally less significant and can be either neglected or included in the static power consumption.

Assumption 3.4-2 We consider the CPU is equivalent in effect to a composite of multiple transistors in dynamic states.

Justification 3.4-2 For every tiny transistor in the CPU, we focus on its effect of storing charges, and thus see it as a tiny capacitor. Thus, the CPU can be modeled as a composite of multiple transistors in dynamic states.

Assumption 3.4-3 The dynamic power consumption is proportional to the voltage squared and frequency.

Justification 3.4-3 According to Assumption 3.4-2, it is very obvious that $P_{\text{dynamic}} \propto U^2$. Furthermore, in unit time, the number of cycles of capacitors being charged and discharged is proportional to the frequency f . Thus, we also have $P_{\text{dynamic}} \propto f$.

Assumption 3.4-4 The frequency is proportional to a power function of the voltage.

Justification 3.4-4 The frequency of a CPU is determined by the clock speed, which is influenced by the voltage supplied to the CPU. By approximating using Taylor expansion, we may assume that $f \propto U^\mu$ within small variations.

Assumption 3.4-5 The formulas for GPU power consumption are similar to those for CPU.

Justification 3.4-5 CPU and GPU share similar architectures and operational principles. Therefore, the formulas for them can be considered similar in format, with differences in constants.

3.4.1 CPU

$$P_{\text{CPU}} = P_{\text{static}} + P_{\text{dynamic}} \quad (15)$$

According to Assumption 3.4-3, in unit time, the dynamic power consumption is proportional to $U^2 \cdot f$, i.e.

$$P_{\text{dynamic}} = \lambda U^2 \cdot f \quad (16)$$

where f is related to U . Suppose

$$f = bU^\mu \quad (17)$$

and the CPU utility is α , then

$$P_{\text{dynamic}} = \alpha U^{2+\mu} b \lambda \quad (18)$$

Citing the works from previous scholars^[4], we have the static power consumption

$$P_{\text{static}} = U \left(c_1 T^2 e^{c_2/T} + I_{\text{gate}} \right) \quad (19)$$

where c_1, c_2 are constants related to the specific CPU, and I_{gate} is the leakage current of the transistor gates.

3.4.2 GPU

Similar to CPU, we have

$$P_{\text{GPU}} = P_{\text{static}} + P_{\text{dynamic}} \quad (20)$$

$$P_{\text{dynamic}} = \alpha' U_{\text{GPU}}^{2+\mu'} b' \lambda' \quad (21)$$

$$P_{\text{static}} = U_{\text{GPU}} \left(c_3 T^2 e^{c_4/T} + I'_{\text{gate}} \right) \quad (22)$$

3.5 Speakers

Citing the works from previous scholars^[5], we have

$$U_{\text{in,req}}(L_c) = 10^{\Delta L/20} U_{\text{ref}} \quad (23)$$

where $U_{\text{in,req}}$ is the required input voltage, L_c is the volume level chosen by the user, $\Delta L = L_c - L_{\text{ref}}$ is the difference between L_c and the reference volume level, and U_{ref} is the reference input voltage.

The reference level is measured under the same circumstances, and vary accordingly with the exact speakers.

And by the definition of power, the power consumption of the speaker is

$$P_{\text{speaker}} = \Re Z(f) \cdot \frac{U_{\text{in,req}}(f)^2}{|Z(f)|^2} \quad (24)$$

where f is the frequency (related to time t), $Z(f) \in \mathbb{C}$ is the complex impedance of the speaker coil, symbol $\Re$ denotes the real part.

3.6 Peripherals

Assumption 3.6-1 We only consider the power consumption of camera and microphone.

Justification 3.6-1 Camera and microphone are the most commonly used peripherals in smartphones, and they significantly contribute to power consumption. Other peripherals, such as sensors or haptic feedback, generally consume less power and can be considered negligible.

Assumption 3.6-2 The power consumption of camera is proportional to the number of pixels and the frame rate.

Justification 3.6-2 The number of pixels determines the amount of data that needs to be processed and transmitted, while the frame rate affects the frequency of data capture and processing. Both factors directly influence the power consumption of the camera.

Assumption 3.6-3 The power consumption of microphone is proportional to the sampling rate and bit rate.

Justification 3.6-3 The sampling rate determines how frequently audio samples are captured, while the bit rate affects the amount of data that needs to be processed and transmitted. Both factors directly influence the power consumption of the microphone.

3.6.1 Camera

According to Assumption 3.6-2, we have

$$P_{\text{camera}} = \gamma \cdot N_{\text{pixels}} \cdot f_{\text{frame}} + P_{\text{default,camera}} \quad (25)$$

3.6.2 Microphone

According to Assumption 3.6-3, we have

$$P_{\text{microphone}} = \delta \cdot f_{\text{sample}} b_{\text{bitrate}} + P_{\text{default,microphone}} \quad (26)$$

3.7 Thermal Effects

Assumption 3.7-1 We only consider the heat generated by the processing units and the battery internal resistance.

Justification 3.7-1 The processing units and the battery internal resistance are the main sources of heat generation in a smartphone. Other components, such as the screen and network modules, generate relatively less heat and can be neglected for simplicity.

Assumption 3.7-2 We consider four key temperatures: the temperature of the processing units, the temperature of the battery, the overall phone environment temperature, and the external environment temperature.

Justification 3.7-2 These four temperatures are crucial in understanding the thermal dynamics of the smartphone. The processing units and battery temperatures directly affect their performance and longevity, while the overall phone environment temperature influences user comfort and device safety, and reflects the working condition of electronic components. The external environment temperature serves as a boundary condition for heat transfer analysis.

Assumption 3.7-3 We see the processing units as pure resistances.

Justification 3.7-3 There is no other form of energy conversion in the processing units except for heat generation due to resistance.

According to Joule's law and Assumption 3.7-3, the heat generated by the units can be described as

$$q_{P.U.} = \frac{dQ_{P.U.}}{dt} = U_{P.U.} I_{P.U.} \quad (27)$$

$$q_{battery} = I_{battery}^2 r_i \quad (28)$$

then we may use the Fourier's law to analyze the heat transfer. Suppose the effective distance between the heat source and the other components is d , and the effective area of heat transfer is A , then we have

$$q_{disp} = k_{P.U.} A_{P.U.} \frac{T_{P.U.} - T_{env}}{d} + k_{battery} A_{battery} \frac{T_{battery} - T_{env}}{d} \quad (29)$$

where k is the thermal conductivity of the material between the source and the overall phone environment, T_{source} is the temperature of the heat source, and T_{env} is the temperature of the phone environment, where all other components are in direct contact with.

Then, for the thermal sources, we have

$$q_{P.U.} - k_{P.U.} A_{P.U.} \frac{T_{P.U.} - T_{env}}{d} = c_{P.U.} m_{P.U.} \frac{dT_{P.U.}}{dt} \quad (30)$$

$$q_{battery} - k_{battery} A_{battery} \frac{T_{battery} - T_{env}}{d} = c_{battery} m_{battery} \frac{dT_{battery}}{dt} \quad (31)$$

where c is the specific heat capacity of the processing unit material, and m is the mass of the processing unit. For the overall phone environment, we have

$$q_{disp} - q_{ext} = c_{env} m_{env} \frac{dT_{env}}{dt} \quad (32)$$

where c_{env} is the overall heat capacity of the phone environment, and q_{ext} is the heat loss to the external environment, which can be modeled using Newton's law of cooling

$$q_{ext} = h A_{phone} \frac{d(T_{env} - T_{ext})}{dt} = h A_{phone} \frac{dT_{env}}{dt} \quad (33)$$

where h is the overall heat transfer coefficient, $A_{phone} = 2A_{screen} \approx A_{P.U.} + A_{battery}$ is the surface area of the phone, and T_{ext} is the temperature of the external environment, which we see as a constant.

3.8 Battery Degradation

Assumption 3.8-1 The capacity degradation of the battery is mainly caused by the loss of lithium ions in the electrolyte during charge-discharge cycles.

Justification 3.8-1 During the charge-discharge cycles, some lithium ions may become trapped in the electrode materials or form solid electrolyte interphase (SEI) layers, leading to a decrease in the number of active lithium ions available for intercalation and de-intercalation. This loss of lithium ions results in a reduction of the battery's overall capacity over time.

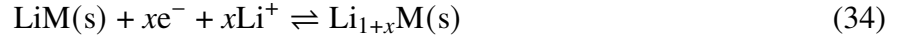
Assumption 3.8-2 We only consider complete draining cycles.

Justification 3.8-2 Complete draining cycles provide a standardized way to measure battery degradation. Moreover, an incomplete draining cycle can be seen as one battery with complete drainage and another battery with no drainage.

Assumption 3.8-3 The loss ratio of lithium ions in the electrolyte during charging and discharging are independent.

Justification 3.8-3 The chambers of charging and discharging reactions are separated, and the corresponding reactions can thus be considered independent.

Assumption 3.8-4 We only consider a generalized reaction



where M is the electrode material.

Justification 3.8-4 The equations for different reactions is only different by constants, which means one can multiply a constant factor to adjust the model for specific reactions. Thus the overall form of the model remains unchanged.

Assumption 3.8-5 We consider the mass of electrolyte and the anode and cathode remain constant during charge-discharge cycles.

Justification 3.8-5 Under normal circumstances, every charge-discharge cycle is very close to a complete reaction, thus the mass of electrolyte and the anode and cathode remain nearly constant.

3.8.1 Charge-Discharge Cycles

By Nernst equation, we have

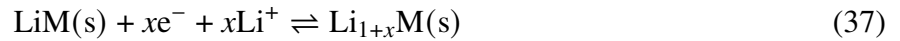
$$\mathcal{E} = \mathcal{E}^\circ - \frac{RT}{nF} \log Q \quad (35)$$

where $\mathcal{E}$ is the electromotive force, $\mathcal{E}^\circ$ is the standard electromotive force (magnitude 3.0401 V for the pair Li/Li⁺), R is the universal gas constant, T is the temperature in Kelvin, n is the number of moles of electrons transferred in the reaction, F is the Faraday constant, and Q is the reaction quotient.

Now we substitute $\mathcal{E} = U_{\min}$, where $U_{\min}$ is the minimum battery output voltage, and we have $Q = K$, the equilibrium constant. Thus,

$$U_{\min} = \mathcal{E}^\circ - \frac{RT}{nF} \log K \Rightarrow K = \exp \frac{nF(\mathcal{E}^\circ - U_{\min})}{RT} \quad (36)$$

We consider the total reaction



and the forward reaction (charging state), we have the equilibrium constant

$$K_+ = c^{-x}(\text{Li}^+) = \left(\frac{M(\text{Li}) \cdot V}{m(\text{Li}^+)} \right)^x \quad (38)$$

where V is the volume of the electrolyte. Hence,

$$m(\text{Li}^+) = M(\text{Li}) \cdot V \cdot K_+^{-\frac{1}{x}} = M(\text{Li}) \cdot V \cdot \exp -\frac{nF(\mathcal{E}_+^\circ - U_{\min+})}{xRT} \quad (39)$$

If we define $1 - r_+$ as the loss ratio of lithium ions in the electrolyte, then

$$r_+ = 1 - \frac{m(\text{Li}^+)}{m_0} = 1 - \frac{M(\text{Li}) \cdot V}{m_0} \cdot \exp - \frac{nF(\mathcal{E}_+^\circ - U_{\min+})}{xRT} \quad (40)$$

Now we consider the reverse reaction (discharging state), we have the equilibrium constant

$$K_- = c^x(\text{Li}^+) = \left(\frac{m(\text{Li}^+)}{M(\text{Li}) \cdot V} \right)^x \quad (41)$$

Using similar methods and note that $\mathcal{E}_+^\circ = -\mathcal{E}_-^\circ$, we have

$$r_- = 1 - \frac{M(\text{Li}^+) \cdot V}{m_0} \exp - \frac{nF(U_{\min-} + \mathcal{E}_+^\circ)}{xRT} \quad (42)$$

and the capacity after N cycles is

$$E_N = E_0 r_+^N r_-^N \quad (43)$$

3.8.2 Internal Resistance

Citing the works from previous scholars^[6], we may use a polynomial to model the internal resistance of the battery

$$r_i = \left(x_1 + x_2(\text{SOC}) + x_3(\text{SOC})^2 + x_4(\text{SOC})^3 \right) x_5 \varphi_N \exp \frac{x_6}{T + x_7} \quad (44)$$

where x_i are constants related to the specific battery, and $\varphi_N = \varphi_0^N$ (we assume) is a coefficient related to the number of cycles.

3.8.3 Ambient Temperature

The ambient temperature affects the battery degradation in two main ways: it influences the rate of chemical reactions within the battery and affects the internal resistance of the battery.

According to the Arrhenius equation, the rate constant k of a chemical reaction is given by

$$k = A \exp - \frac{E_a}{RT} \quad (45)$$

where A is the pre-exponential factor, E_a is the activation energy. Thus, higher ambient temperatures lead to increased reaction rates, which can accelerate the degradation processes within the battery.

As for the internal resistance, we have already modeled it in Eq.(40), where an increase in temperature generally leads to a decrease in internal resistance, which can improve battery performance but may also contribute to faster degradation due to increased reaction rates.

Furthermore, if we consider the brownian motion of lithium ions, we have the diffusion coefficient

$$D = \frac{k_B T}{6\pi\eta r} \quad (46)$$

where η is the viscosity of the electrolyte, and r is the radius of the lithium ion. Higher temperatures lead to increased diffusion coefficients, which can enhance the mobility of lithium ions within the battery. This increased mobility can improve battery performance but may also contribute to faster degradation due to increased reaction rates.

Last, extreme temperatures (both high and low) can cause the SEI layer to collapse or become unstable, leading to increased lithium ion loss and accelerated capacity fade.

4 Numerical Simulation and Results

4.1 Parameter Values for Short-Term Simulation

The parameters we choose are listed below. Some values are estimated based on typical smartphone specifications, while others are derived from empirical data or literature. Those without specified units are either not convenient to express in SI (indicated with N/A) or based on SI conventions (indicated with a space) or with no unit in itself (indicated with a dash).

4.1.1 Screen

Parameter	Value	Unit (Only those not in SI)
E_{a1}	0.7	eV
E_{a2}	0.2	eV
P_R	100	%
P_G	100	%
P_B	100	%
(Br)	75	%
a_R	3.399 6	N/A
a_G	7.050 3	N/A
a_B	19.508 1	N/A
$a_0 a_R$	242.318 0	N/A
$a_0 a_G$	293.699 9	N/A
$a_0 a_B$	294.013 9	N/A
L_0	0.01	N/A
ω	$\pi/12$	rad/hour
δ	0	rad
L_{base}	0.01	N/A

4.1.2 Network

Parameter	Value	Unit (Only those not in SI)
$P_{network}$	0.2	

4.1.3 Processing Units

Parameter	Value	Unit (Only those not in SI)
α	0.5	N/A
α'	0.5	N/A
b	1.0	N/A
b'	1.0	N/A
μ	1	-
μ'	1	-
λ	2	N/A
λ'	2.5	N/A
c_1	0.255 0	N/A
c_2	-374 0	N/A
c_3	0.256 1	N/A
c_4	-374 0	N/A
I_{gate}	1×10^{-7}	
I'_{gate}	8.6×10^{-8}	

4.1.4 Speaker and Peripherals

Parameter	Value	Unit (Only those not in SI)
P_{speaker}	0.3	
P_{camera}	0.02	
$P_{\text{microphone}}$	0.1	

4.1.5 Thermal Effects

Parameter	Value	Unit (Only those not in SI)
$U_{\text{P.U.}}$	3.7	
$I_{\text{P.U.}}$	2.0	
I_{battery}	0.5	
d	0.005	
$c_{\text{P.U.}}$	712	
$m_{\text{P.U.}}$	0.005	
$k_{\text{P.U.}}$	1	N/A
c_{battery}	900	
m_{battery}	0.12	
k_{battery}	2.5	N/A
m_{env}	0.06	
A_{phone}	$2 \times 0.011\ 86$	
$A_{\text{P.U.}}$	$2 \times 0.001\ 75$	
A_{battery}	$2 \times 0.000\ 06$	

4.1.6 Miscellaneous

For the battery internal resistance, we use the following data under 0.75C conditions.

Discharge Rate \ Index	1	2	3	4	5	6	7
0.75C	4.834 7	-0.243 7	-0.375 1	-0.162 8	6.057 7	2.692 1	-0.815 9
1.75C	4.236 2	-0.276 0	-0.624 8	-0.024 0	19.906 8	2.725 3	0.182 2
2.75C	3.382 4	-1.097 3	-0.366 8	0.712 6	23.528 7	2.666 8	0.094 0

We set the initial temperature to be 298 K.

The sources of the data above: see works [7] to [13].

4.2 Short-Term Simulation

Here we simulate the battery drainage over one cycle. We monitored the parameter (SOC) and temperatures of different entities during the period. The values are plotted below.

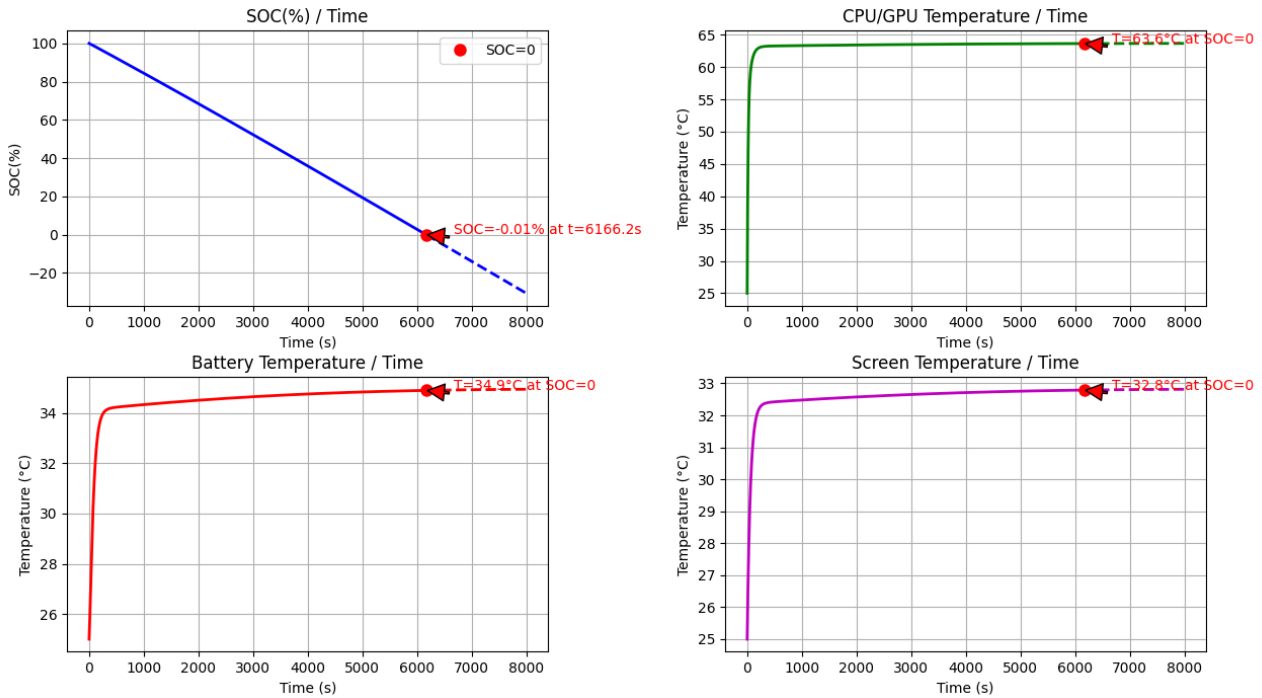


Figure 2: (SOC) over Time and Temperature monitoring

We configured a high load scenario, including high screen brightness, intensive CPU and GPU usages, continuous network activity, and high volume level. As shown in the figure, the battery drains from 100% to 0% in approximately 2 hours. The temperatures of the processing units and battery rise significantly during this period, while the overall phone environment temperature remains relatively stable due to heat dissipation.

4.3 Long-Term Simulation

Here we simulate the battery degradation over 100 complete charge-discharge cycles. The values of parameters used in Eq.(40) and Eq.(42) are as follows (In consideration of the usual charging voltage 5 V, we set $U_{\min-} + U_{\min+} = 5$ V, and other non-scientific constants are chosen by real-life experience):

Parameter	Value
$M(\text{Li})$	7 g/mol
m_0	50 g
V	$2.5 \times 10^{-7} \text{ m}^3$
$\mathcal{E}_+^\circ$	3.0401 V
$U_{\text{min}+}$	1.3 V
$U_{\text{min}-}$	3.7 V
R	8.314 J/(mol·K)
F	96485 C/mol
T	298 K
n	0.35 mol
x	1

where the $\mathcal{E}_+^\circ$ value is from the CRC Handbook of Chemistry and Physics (see [11]).

We calculate

$$r_+ \approx 0.997\,179\,320\,7 \quad (47)$$

$$r_- \approx 1 \quad (48)$$

Thus, we plot the capacity degradation over 500 complete charge-discharge cycles as follows.

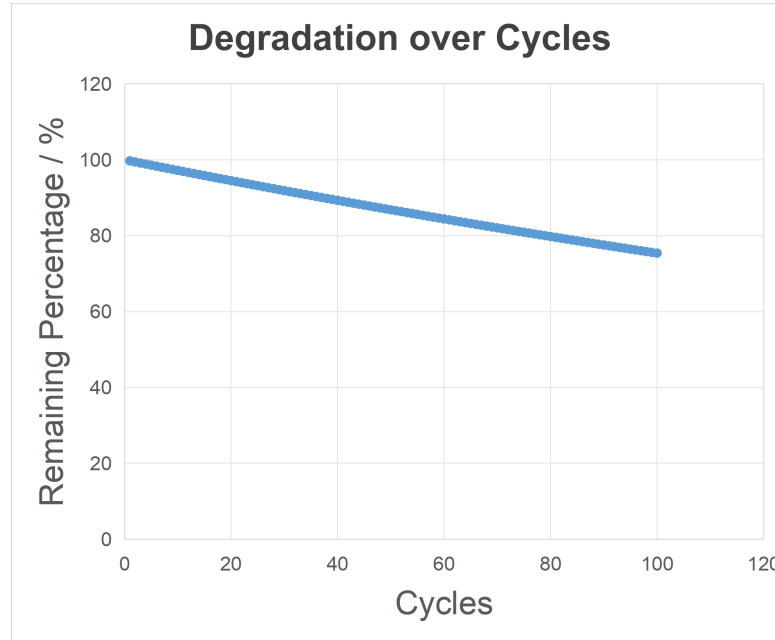


Figure 3: Capacity Degradation over 100 Complete Charge-Discharge Cycles

After 100 complete charge-discharge cycles, the battery capacity drops to approximately 75.392% of its original capacity.

4.3.1 Preliminary Validation

We compared our simulated total power with the power measured in experiment. The results are as follows.

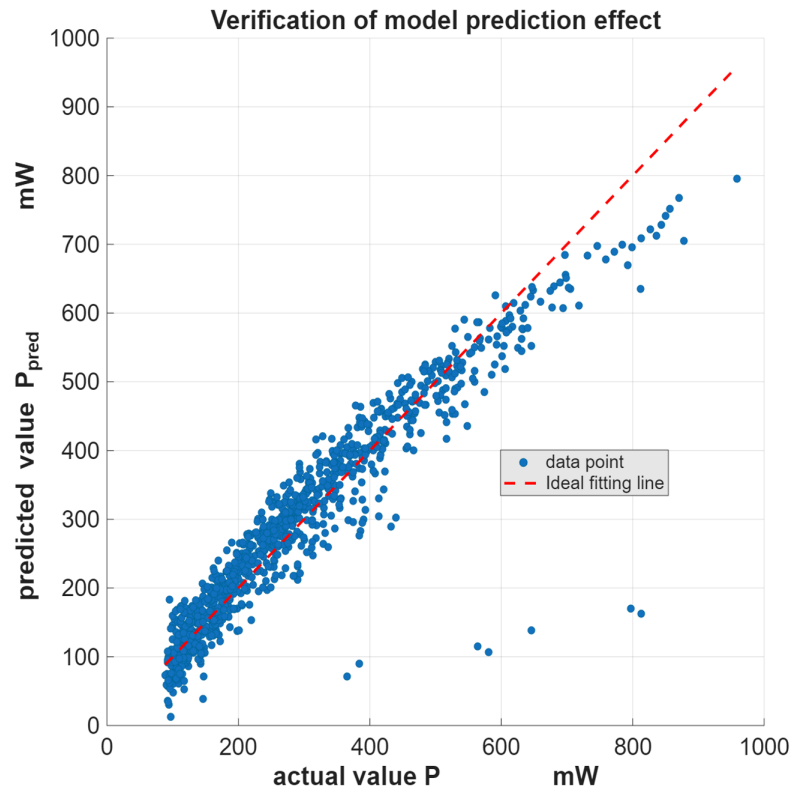


Figure 4: Comparison between Simulated Power and Measured Power

By calculation, we have a correlation coefficient of 0.938 2, indicating a strong positive correlation between the simulated and measured power values. This preliminary validation suggests that our model is to some extent accurate in predicting smartphone battery consumption.

5 Factor and Sensitivity Analysis

In this section, we analyze the influence of different factors. The method is to completely remove one factor out of the model, and simulate the battery drainage over one cycle again. Then the results are compared with the original simulation under 298 K.

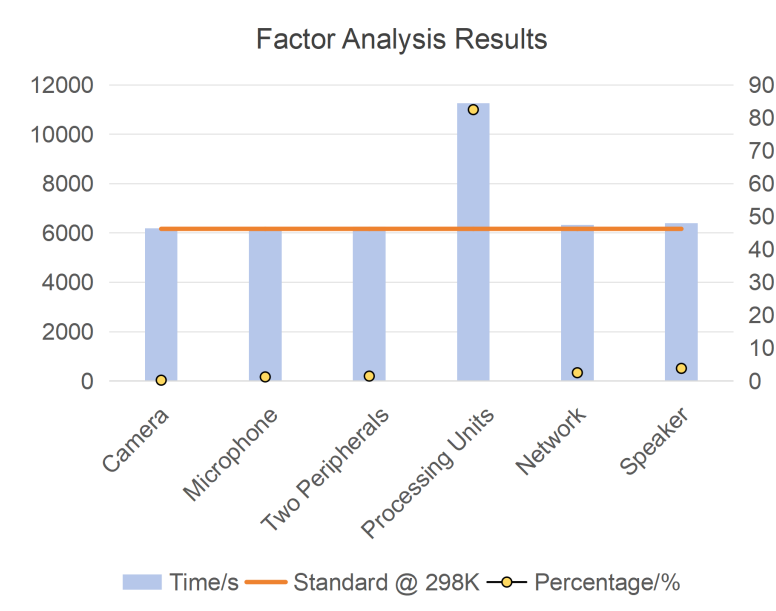


Figure 5: Factor Analysis Results

The relevant data are listed below.

Factor Removed	Shutdown Time	Increase in Shutdown Time (%)
None	6 166.2	0.000 0
Camera	6 181.4	0.246 5
Microphone	6 243.0	1.245 5
Two Preipherals	6 258.2	1.492 0
Processing Units	11 252.3	82.483 5
Network	6 321.4	2.516 9
Speaker	6 403.0	3.840 3

From the results above one can clearly see that, apart from the processing units, the speaker factor has the most significant impact on battery drainage patterns, which comes quite unprecedented. After that is the network factor. The two peripherals have relatively minor effects on battery drainage patterns.

Apart from the factor analysis above, we also perform a sensitivity analysis on ambient temperature. The results are shown below.

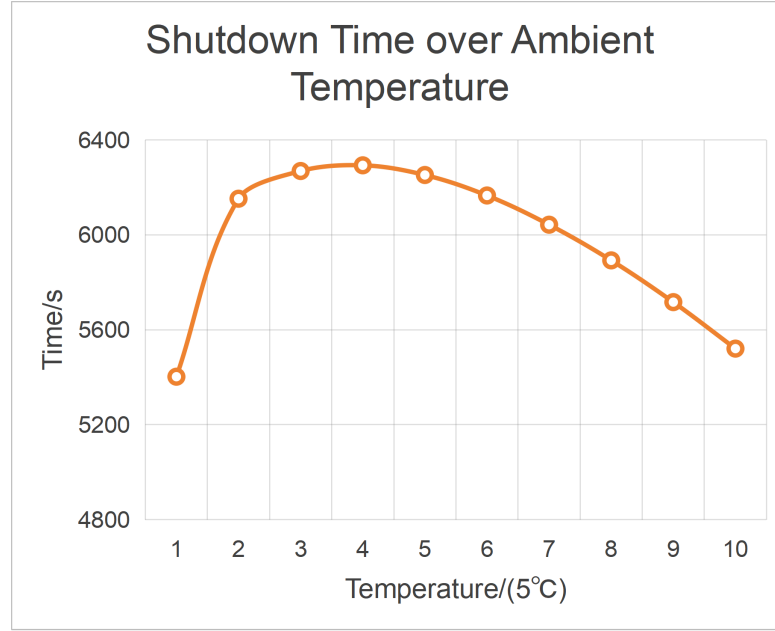


Figure 6: Sensitivity Analysis on Ambient Temperature

The relevant data are listed below.

Ambient Temperature (°C)	Shutdown Time
0	5 402.2
5	6 152.6
10	6 269.4
15	6 293.4
20	6 252.6
25	6 166.2
30	6 043.0
35	5 892.6
40	5 716.6
45	5 520.6

One can see, that under extreme temperatures (both high and low), the battery drains much faster. The optimal ambient temperature for battery life is somewhere around 20°C.

In terms of other parameters, one can clearly see from a mathematical point of view, that all equations regarding exponential functions are more sensitive to changes. Thus, we conclude that parameters in those equations are more sensitive. This includes parameters in Eq.(1), Eq.(4), Eq.(7), Eq.(20), Eq.(40), and Eq.(42): E_{a1} , E_{a2} , c_2 , c_4 , $\mathcal{E}_+^o$, $U_{\min+}$, $U_{\min-}$, n , x . As for other parameters, they are relatively less sensitive.

6 Model Strengths and Weaknesses

6.1 Strengths

1. **Comprehensive Consideration of Factors:** Our model takes into account a wide range of factors that influence smartphone battery consumption, including screen usage, processing units, network activity, speakers, peripherals, thermal effects, and battery degradation. This comprehensive approach allows for a more accurate representation of real-world battery behavior.
2. **Theoretical Foundation:** The model is built upon established physical and chemical principles, such as the Arrhenius equation, Nernst equation, and Fourier's law. This theoretical foundation enhances the credibility and reliability of the model.
3. **Incorporation of Empirical Data:** Where necessary, the model incorporates empirical data and findings from previous scholarly works. This integration of empirical evidence helps to validate the theoretical aspects of the model and improve its accuracy.

6.2 Weaknesses

1. **Data Limitations:** The accuracy of our model is heavily dependent on the quality and quantity of the data used for calibration and validation. Limited or biased data can lead to inaccuracies in the model's predictions.
2. **Simplified Assumptions:** To make the model tractable, we have made several simplifying assumptions, such as neglecting certain minor factors affecting battery performance. These assumptions may limit the model's applicability in real-world scenarios where these factors play a significant role.
3. **Dynamic Usage Patterns:** The model may not fully capture the dynamic and complex usage patterns of smartphone users, which can lead to discrepancies between predicted and actual battery performance.
4. **Theoretical Limitations:** While we have incorporated various physical and chemical principles into our model, there may still be theoretical limitations in our understanding of battery behavior that could affect the model's accuracy, due to our limited knowledge in chemistry and physics.

7 Practical Recommendations

7.1 OEMs

1. Improvements in battery chemistry can significantly enhance battery life.
2. Optimize CPU and GPU architectures for energy efficiency. For example, implementing performance and efficient core designs, dynamic voltage distribution between cores, even between CPU and GPU, can help reduce power consumption during high-load tasks.

3. Reuse heat generated by components. One is easy to think of the Thomson effect and the Seebeck effect, which uses semi-conductors to convert temperature differences directly into electricity.
4. Update device firmwares and drivers. Ensure that the latest firmwares and drivers utilize the unique features of the hardware to cut power consumption.

7.2 Operating System Developers

Here we mostly consider the network factor, which leads us to the following recommendations.

1. Optimize background app network usage. If not necessary, postpone background network activities.
2. Provide user controls for network usage. Allow users to customize preferences for network-intensive activities, such as limiting data usage for specific apps when the battery is low.
3. Optimize data transmission protocols. Use more efficient data compression and transmission techniques to reduce the amount of data sent and received.

Apart from network, OS developers can also optimize CPU and GPU scheduling to reduce power consumption during high-load tasks.

7.3 App Developers

1. Reduce usages in peripherals. For example, avoid unnecessary camera and microphone usage in the background.
2. Optimize algorithms. For example, some algorithms can be replaced with less accurate ones (still within acceptable error margins) that consume less power.
3. Reduce network usage. Minimize data transmission by optimizing data synchronization and caching strategies. Moreover, cut all hidden uncensored and unnotified data collection and transmission.
4. Categorize computation loads. One may categorize loads into those suitable for CPU and those for GPU, and allocate them accordingly to optimize power consumption.

7.4 End Users

1. Cut all unnecessary background processes. This may be done by system settings, avoiding installing unnecessary apps, and avoiding authorizing unnecessary permissions.
2. Adjust screen brightness and timeout settings. Lowering screen brightness and reducing the screen timeout duration can significantly decrease power consumption.
3. Use power-saving modes. Most smartphones have built-in power-saving modes that limit background activities and reduce performance to extend battery life.

4. Avoid extreme temperatures. Keep the smartphone within recommended temperature ranges to prevent accelerated battery degradation.
5. Optimize charging habits. Avoid frequent complete discharges and overcharging to prolong battery lifespan.

8 Works Cited

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- [11] Haynes, W. M. et al, "CRC Handbook of Chemistry and Physics", 97th Edition, CRC Press, Boca Raton, FL, 2016-2017.

Other Online Resources Used:

- [12] "Specs of *Google Pixel 8*" (translated) <https://detail.zol.com.cn/1773/1772430/param.shtml>
- [13] Guegain, E. et al. "AndroWatts: Unpacking the Power Consumption of Mobile Device's Components". <https://zenodo.org/records/14314943>

9 Report on Use of AI

1. Baidu Fanyi, *Baidu Translate* (Dec 9, 2025 version)
Used for translating English terminologies into Chinese, and vice versa.
2. GitHub *CoPilot* (Latest version)
Auto-completion for code used in preparing our models.